B. Tech Summer / ~~Winter~~ - 2024
(ESE / ~~Make-up~~ / Supplementary) Examination

Branch: Computer Science & Engineering /Artificial Intelligence **Sem:- IV**
Subject: Formal Language Automata Theory / Theory of Computation
Subject Code: CS4T001/AI4T001
Marks: 100 **Date of Exam: 11/06/2024** **Time: 3 Hrs.**

Instructions to the students:

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Assume suitable data whenever necessary.**
4. **Illustrate your answer whenever necessary with the help of neat sketches.**
5. **Use of Programmable Calculator is prohibited.**
6. **Don't use red pen for writing the answers.**

COs: Students should be able to:

1. Define the mathematical principles behind theoretical computer science.
2. Differentiate and give examples for the different types of automata like finite automata, push down automata, linear bounded automata and Turing machine.
3. Correlate the different types of automata to real world applications.
4. Choose and design appropriate automata for the different requirements outlined by theoretical computer science.
5. Identify the different computational problems and their associated complexity.

SECTION A

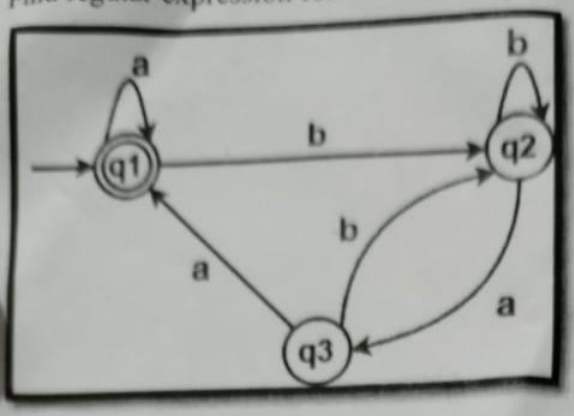
1. **All Questions are compulsory.**
2. **Answer in one/two word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define the string.	L1/CO1	2
Q.2	Draw a finite automata model.	L1/CO1	2
Q.3	Rewrite the meant of epsilon transition.	L2/CO1	2
Q.4	Restate the role of Read/ Write head in the Turing machine.	L2/CO2	2
Q.5	State the statement of Arden's theorem.	L2/CO2	2
Q.6	Enlist the applications of Context free languages.	L1/CO2	2
Q.7	Discuss derivation and derivation/parse tree in short.	L2/CO1	2
Q.8	Define Ambiguous grammar.	L1/CO1	2
Q.9	Write short note on Church's Hypothesis.	L3/CO2	2
Q.10	State the meaning of Unit productions.	L1/CO1	2
Q.11	Identify the tuple formation of Turing Machine.	L1/CO1	2
Q.12	Define P and NP problem.	L2/CO1	

No.

Q.13 Differentiate Moore and Mealy machines. L3/CO

Q.14 Find regular expression for the following DFA.



Q.15 Eliminate useless symbols and productions, unit productions and null productions from the following grammar. L2/CO

S → a | aA | B | C

A → aB | ϵ

B → Aa

C → cCD

D → ddd

universal
random

Q.16 Analyze the two normal forms of Context Free Grammar. L4/C

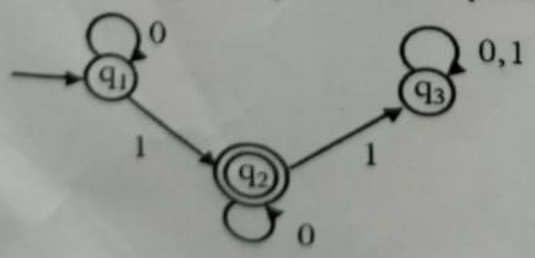
Q.17 Illustrate Post Correspondence Problem with example. L3/C

Q.18 Define formally Type 0, Type 1, Type 2 and Type 3 grammar. Show the corresponding automata for each class L3/C

SECTION C

1. **Solve any FOUR Questions.**

Q. No.	Questions	Level Map
Q.19	Design a Moore machine to determine the residue of mod 3 of the input treated as a binary string.	L4/C
Q.20	Construct a regular expression corresponding to the automata given below	L5/C



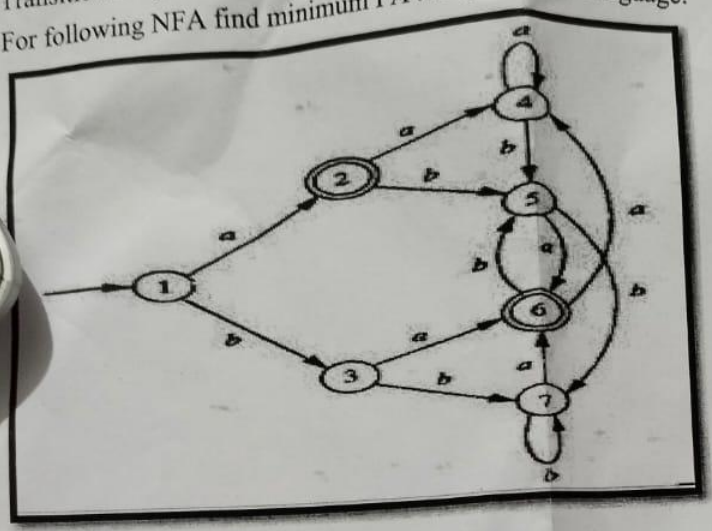
Q.21 Show that CFG

S → a | Sa | bSS | SSb | SbS is ambiguous.

Q.22 Design regular expression for all the strings having "110" as a substring L3/C

Q.23 Construct a TM for the language $L = \{0^n1^n2^n\}$ where $n \geq 1$ Also Draw the Transition Diagram. L4/CO5 7

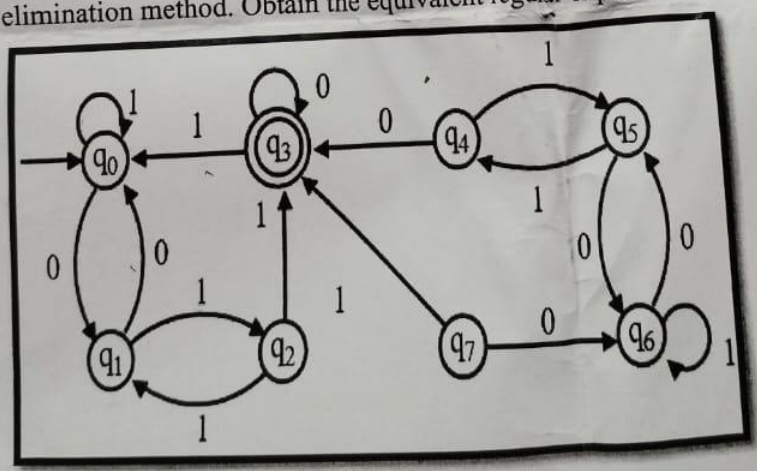
Q.24 For following NFA find minimum FA accepting same language: L5/CO3 7



SECTION D

1. **Solve any TWO Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Minimize the following DFA using table finite automation using state elimination method. Obtain the equivalent regular expression.	L4/CO2	12



Q.26	Define Turing machine with the various types of Turing machines. Explain each type in brief.	L6/CO5	12
Q.27	Construct a NPDA for the language L= {wwR : w->{0,1}*}	L5/CO4	12